

Equilibrium-Generated Regimes and Constitutive Regime Formation

A Post-Equilibrium Classification of Dynamic Games

Marcus Hermansson

Independent researcher

Correspondence: <https://hmwh.se/>

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A certified dynamic equilibrium induces a state process, but the equilibrium concept alone does not classify the persistent or asymptotic regime generated by that process. This paper introduces an equilibrium-generated regime (EGR): a selected equilibrium policy together with a predeclared regime region, initial basin, persistence criterion, descriptor, limiting occupation law, and convergence mode. A narrower *Permansson regime* is an EGR for which a fixed admissible policy intervention changes a defining regime property. The framework is therefore not a new equilibrium solution concept; it is a post-equilibrium classification conditional on an independently established equilibrium certificate.

The paper develops a multi-axis regime profile separating asymptotic form, strategic-mode locking, constitutive dependence, resistance dynamics, and opponent-mediated cross-effects. It distinguishes material cross-generation from reinforcement of deviation resistance and embeds both in a signed feedback architecture that also permits asymmetric endogenous self-limitation. A canonical two-player engagement game establishes a stationary Markov-perfect equilibrium, exact invariance conditions, Perron compositional convergence, and uniform constitutive dependence of balanced growth on mutual engagement. A nonlinear mobilisation–cohesion extension nests the linear benchmark and provides a minimal representation of state-dependent mobilisation and internal limitation, without claiming general equilibrium, recurrence, or stability results. The institutional module separates belief, latent recognition, observed action, and operative classification; a reduced-form claim-or-wait benchmark derives order-sensitive branch incentives and clarifies the distinction between first-mover advantage and first-state-transition advantage. The contribution is a formally specified,

intervention-relative framework for classifying equilibrium-generated regimes and the feedback structures that sustain, limit, or redirect them.

Keywords: stochastic games; Markov-perfect equilibrium; equilibrium-generated regime; constitutive intervention; invariant region; opponent-mediated dynamics; feedback architecture; endogenous self-limitation; balanced growth; institutional recognition; pre-emption.

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1. Introduction

1.1. The question after equilibrium

A stochastic game combines strategic choice with a payoff-relevant state and an action-dependent transition law (Shapley 1953; Başar and Olsder 1999). A Markov-perfect equilibrium (MPE) certifies sequentially rational state-dependent behaviour (Maskin and Tirole 2001). Once such a policy has been identified, however, a distinct set of questions remains. Does equilibrium play preserve a meaningful state region? Does the induced process converge, recur, contract, collapse, or grow? Does the same strategic mode persist as the state changes? Does one actor's equilibrium action generate resources for another actor, or does it also increase the other's incentive to remain in the same mode? Would the same qualitative regime occur under a predeclared counterfactual intervention on equilibrium behaviour?

These are not alternative equilibrium conditions. They concern the state process generated after a policy and transition law are combined. The analytical target is therefore post-equilibrium: the regime induced by an independently certified equilibrium.

1.2. Contribution

The paper makes five contributions.

First, it defines an *equilibrium-generated regime* (EGR), which combines a selected equilibrium policy with a predeclared region, an initial-state basin, exact persistence, and a stated asymptotic descriptor. The definition distinguishes exact EGR persistence from finite-horizon quasi-regimes and metastable extensions, and a limiting occupation law from an invariant law of a Markov factor.

Second, it introduces an intervention-relative criterion for *constitutive regime formation*. A policy component is constitutive only if a fixed admissible policy intervention changes a predeclared regime property while the action sets, payoffs, and transition kernel remain fixed. Structural and institutional interventions are treated separately and identify constitutive mechanisms rather than constitutive equilibrium action.

Third, it develops a *multi-axis regime profile* separating asymptotic form, strategic-mode structure, constitutive dependence, resistance dynamics, opponent-mediated cross-effects, and optional welfare annotations. These labels are not assumed to be mutually independent: the point is to prevent distinct properties from being collapsed into one adjective.

Fourth, it separates *cross-generation* from *cross-incentive reinforcement* and places both inside a signed feedback architecture. The framework can therefore represent not only positive opponent-mediated effects but also asymmetric self-limitation, counter-mobilisation, and restorative feedback. The linear canonical model isolates reciprocal state generation; a nonlinear extension shows how mobilisation and cohesion can enter the state without altering the proved linear benchmark.

Fifth, it develops a recognition architecture distinguishing belief, latent recognition, observable institutional action, and operative classification. A reduced-form claim-or-wait benchmark shows how an order premium can move claim incentives across best-response boundaries when operative classification changes continuation values. The same architecture identifies first-state-transition

advantage and causal-origin capture as institutionally meaningful subtypes, while leaving a full dynamic recognition EGR to future work.

1.3. Scope

The formal core assumes a selected pure stationary MPE in a discounted stochastic game with a fully observed payoff-relevant state, finite action sets, bounded payoffs, and a measurable transition kernel. The paper does not prove general existence of pure MPEs. All regime classifications are relative to both the selected equilibrium and the analyst's predeclared regime specification. A game with multiple equilibria may therefore support multiple EGRs.

The framework is called *Permansson Regime Theory* (PRT), but PRT is not an alternative solution concept. The narrower named object is a *Permansson regime*. The proved core comprises the exact EGR architecture, the linear canonical game, the reduced-form claim game, and two limited nonlinear results: state-space preservation and exact nesting of the linear benchmark. The modular state interpretation, multi-axis profile, signed feedback graph, distributed-recognition representation, and causal-origin subtype are definitions, diagnostics, or research objects; they do not add general equilibrium or stability theorems.

2. Related Literature and Contribution Boundary

2.1. Dynamic equilibrium and induced state processes

Stochastic and dynamic games already permit present actions to alter future states, feasible actions, payoffs, and information (Shapley 1953; Başar and Olsder 1999; Maskin and Tirole 2001). PRT therefore does not claim that Nash or MPE presupposes a static world. The distinction is one of analytical target: an equilibrium concept certifies a policy, whereas an EGR classifies the persistent and asymptotic process induced by that certified policy.

Work on Markov distributional equilibria explicitly studies equilibrium transition paths and the invariant distributions to which they converge (Balbus et al. 2022). Related work derives probabilistic economic equilibrium as an invariant distribution generated by purposive dynamic behaviour (Jacobo 2026), while subjective-equilibrium models study closed-loop environments that players incorrectly treat as exogenous (Arslan and Yüksel 2023). Markov-chain theory, stochastic stability, and viability theory provide established tools for invariant sets, recurrent classes, perturbations, and long-run laws (Aubin 1991; Young 1993; Meyn and Tweedie 2009). PRT does not claim priority for equilibrium-induced processes or invariant order. Its narrower object combines an independently certified equilibrium with a predeclared regime specification and an intervention-relative constitutive diagnostic.

2.2. Strategy–environment feedback and endogenous games

Evolutionary games with environmental feedback study reciprocal coupling between behaviour and an evolving environment, including oscillation, bistability, tipping, and long-run composition

(Weitz et al. 2016; Tilman et al. 2020; Ito and Yamamichi 2024). Endogenous-institution models study institutions that alter the conditions of their own reproduction (Greif and Laitin 2004). Endogenous-game and mode-switching models allow actors to change feasible interactions or move between payoff regimes (Conley and Neilson 2009; Frey and Atkisson 2020; Dawid and Gezer 2022). Concurrent work studies intervention-driven equilibrium transition under status-quo inertia and endogenous feasibility constraints (Eshaghi Gordji et al. 2026; Omolere 2026). Those papers alter equilibrium selection, feasibility, or the game itself. PRT instead conditions on a separately certified dynamic equilibrium and uses an admissible policy intervention as a diagnostic of whether equilibrium behaviour is constitutive of the induced regime classification.

2.3. Action–reaction systems and positive dynamics

The canonical model belongs to the broad tradition of Richardson-type action–reaction systems and positive linear dynamics. Strategic and differential-game formulations of arms races predate the present framework (Brito 1975; Ho and Mookherjee 1975). Perron–Frobenius convergence is standard positive-matrix theory (Bapat and Raghavan 1997; Seneta 2006). The canonical model is consequently an existence construction and fully solved benchmark, not a claim to new linear algebra or a new arms-race equation.

2.4. Pre-emption, recognition, and institutional status

Pre-emption games analyse races for first-mover positions or investment opportunities (Fudenberg and Tirole 1985; Anderson et al. 2010; Thijssen 2010). Recognition and status contests are also established subjects; the phrase “recognition game” has prior use in international-relations scholarship (Ringmar 2002). Roll-call and ideal-point models already distinguish latent preferences from observed institutional acts and warn that agenda, abstention, bundling, and strategic voting complicate inference (Clinton et al. 2004; Bailey et al. 2017; Häge and Hug 2016). The present module therefore does not claim priority for latent-state inference, strategic recognition, or first-mover advantage. Its narrower contribution is to add an operative classification state that changes continuation-game consequences and to connect an order premium to best-response boundaries in a public benchmark game.

2.5. Structural counterfactuals and empirical dynamic games

The constitutive criterion is counterfactual and therefore requires an explicit statement of what an intervention changes and what remains invariant. This is consistent with the structural approach to policy counterfactuals and intervention semantics (Balke and Pearl 1995; Pearl 2009; Wolf and McKay 2023). In empirical dynamic games, equilibrium recovery, payoff identification, information assumptions, and equilibrium selection are substantive econometric problems (Bajari et al. 2007; Pesendorfer and Schmidt-Dengler 2008; Aguirregabiria and Magesan 2020; Bugni and Bunting 2021; Magnolfi and Roncoroni 2023; Dearing and Blevins 2025; Berry and Compiani 2023). PRT does not solve those problems; it states what must be identified before an empirical regime classification is credible.

2.6. Closest neighbours

Table 1. Closest neighbouring literatures and the remaining distinction.

Literature	Existing object	Remaining PRT distinction
Stochastic games	Equilibrium policy and induced state process	A predeclared regime tuple and constitutive policy intervention
Invariant and probabilistic order	Invariant sets, recurrence, occupation laws, equilibrium-induced invariant distributions	Classification remains conditional on a selected equilibrium certificate and declared descriptor
Strategy–environment feedback	Coupled behaviour and environmental state	Separate policy certification, mode resistance, and intervention-relative constitution
Endogenous institutions and game transformation	Institutions, feasibility, or equilibrium selection change endogenously	Post-equilibrium classification of a certified policy-generated regime
Action–reaction models	Reciprocal growth, mobilisation, or dissipation	Separation of state cross-generation, incentive reinforcement, and endogenous limitation
Pre-emption, recognition, and latent-state models	Timing races, status contests, latent preferences, and observed votes	Operative classification, branch consequences, and first-state-transition advantage

The scoped novelty claim is organisational and diagnostic. In the literature searched through 1 August 2026, we did not identify a direct predecessor combining the selected equilibrium certificate, predeclared region and initial basin, exact persistence and asymptotic descriptor, fixed admissible policy intervention, mode-resistance diagnostic, opponent-mediated cross-effects, and signed feedback architecture in one post-equilibrium object. This is not a priority claim for the individual mathematical ingredients, intervention types, feedback mechanisms, or recognition phenomena.

3. Formal Environment

3.1. Primitive game and equilibrium certificate

Consider a discounted stochastic game

$$\Gamma = (N, \mathcal{X}, \{A_i\}_{i \in N}, \{u_i\}_{i \in N}, P, \{\beta_i\}_{i \in N}). \quad (1)$$

Assumption 3.1 (Core environment). (A1) $N = \{1, \dots, n\}$ is finite.

(A2) $\mathcal{X}$ is a Polish space equipped with its Borel σ -algebra and a declared reference measure m .

- (A3) Each player has a finite master action set $\bar{A}_i$, and the nonempty feasible-action correspondence $A_i(x) \subseteq \bar{A}_i$ has a measurable graph.
- (A4) Each stage payoff is bounded and measurable on the joint feasible state-action graph.
- (A5) $P(\mathrm{d}x' \mid x, a)$ is a measurable transition kernel on feasible state-action pairs.
- (A6) $\beta_i \in (0, 1)$ for every player.
- (A7) A measurable pure stationary MPE π^* has been selected.

The last condition is an equilibrium certificate, not an existence claim. A pure stationary profile π induces

$$P^\pi(B \mid x) = P(B \mid x, \pi(x)) \quad (2)$$

for measurable $B \subseteq \mathcal{X}$, and value

$$V_i^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{t=0}^{\infty} \beta_i^t u_i(X_t, \pi(X_t)) \right]. \quad (3)$$

For a one-period unilateral action a_i followed by π , define

$$Q_i^\pi(x, a_i) = u_i(x, a_i, \pi_{-i}(x)) + \beta_i \int V_i^\pi(x') P(\mathrm{d}x' \mid x, a_i, \pi_{-i}(x)). \quad (4)$$

Under Assumption 3.1(A1)–(A6), the one-shot inequalities

$$Q_i^\pi(x, \pi_i(x)) \geq Q_i^\pi(x, a_i) \quad \forall i, x, a_i \in A_i(x) \quad (5)$$

are equivalent to optimality against all admissible unilateral behavioural strategies. Full proof details are given in the technical supplement. A proposed regime region never restricts the deviation set: equilibrium is always evaluated in the complete game.

3.2. Local continuation-action problem

Conditional on π^* , define the local continuation-action problem

$$\mathcal{C}_{\pi^*}(x) = \left(N, \{A_i(x)\}_{i \in N}, \{Q_i^{\pi^*}(x, \cdot)\}_{i \in N} \right). \quad (6)$$

The equilibrium process generates a sequence $\mathcal{C}_{\pi^*}(X_0), \mathcal{C}_{\pi^*}(X_1), \dots$. This is a state-indexed representation of current action incentives under the fixed continuation policy, not the full continuation subgame.

3.3. Modular state architecture

The abstract state can be interpreted as a finite product of payoff-relevant components,

$$X_t = \left(X_t^{\text{mat}}, X_t^{\text{inst}}, X_t^{\text{bel}}, X_t^{\text{net}}, X_t^{\text{impl}}, X_t^{\text{data}} \right), \quad (7)$$

representing, for example, material capability, operative institutional classification, beliefs or latent recognition, interaction networks, cohesion or implementation capacity, and an endogenous data-generating environment. An application includes only the components required to make the selected equilibrium, transition law, and claimed regime mechanism Markovian. A finite product of Polish component spaces is Polish, so this interpretation does not alter assumption 3.1 or the dynamic-programming results.

The modular representation is analytical rather than ontological: it does not imply that every domain contains every component, or that a component can be identified independently from available data. It clarifies that equilibrium behaviour may reconstruct material, institutional, informational, organisational, or data-generating parts of the payoff-relevant state.

4. Equilibrium-Generated Regimes

4.1. Equilibrium-generated process and analyst-relative specification

Given Γ , the selected equilibrium π^* , and an initial law λ_0 , the equilibrium-generated process is the Markov process with initial distribution λ_0 and transition kernel P^{π^*} . An EGR adds an analyst-declared specification. It is therefore relative to

$$(\pi^*, \mathcal{B}, B_0, h, \nu, \mathfrak{c}), \quad (8)$$

not uniquely determined by Γ alone.

Here $\mathcal{B} \subseteq \mathcal{X}$ is a regime region, $B_0 \subseteq \mathcal{B}$ is an initial-state basin, $h : \mathcal{B} \rightarrow \mathcal{Y}$ is a descriptor into a compact metrizable space, ν is a limiting occupation law, and $\mathfrak{c}$ records a mode of convergence. Let

$$\mathcal{D}(B_0) = \{\lambda \in \Delta(\mathcal{X}) : \lambda(B_0) = 1\} \quad (9)$$

be the admissible initial-law class.

Assumption 4.1 (Ex ante non-triviality). The sets $B_0 \subseteq \mathcal{B} \subseteq \mathcal{X}$ are Borel, the descriptor h is Borel measurable, and the region, basin, descriptor, and convergence criterion are specified before the realised trajectory is classified. The basin contains at least two states inducing distinct path laws; $\mathcal{B}$ has nonempty interior or positive m -measure; and $h(\mathcal{B})$ contains at least two points. The same descriptor is used under equilibrium and intervention laws.

These are minimal guardrails, not an identification theorem. Exploratory searches for candidate regions should be conducted on separate simulations or data and confirmed on held-out material.

4.2. Persistence

Define the first exit time

$$\tau_{\mathcal{B}} = \inf\{t \geq 0 : X_t \notin \mathcal{B}\}. \quad (10)$$

The region is *exactly invariant* if

$$P^{\pi^*}(\mathcal{B} \mid x) = 1 \quad \forall x \in \mathcal{B}. \quad (11)$$

This is equivalent to pathwise retention at all dates and to $\mathbb{P}_x^{\pi^*}(\tau_{\mathcal{B}} = \infty) = 1$ for every $x \in \mathcal{B}$.

For $H \in \mathbb{N}$ and $\eta \in [0, 1]$, a region is (H, η) -persistent if

$$\inf_{x \in \mathcal{B}} \mathbb{P}_x^{\pi^*}(\tau_{\mathcal{B}} > H) \geq 1 - \eta. \quad (12)$$

A uniform one-period retention probability of at least $1 - \eta$ implies the finite-horizon lower bound $(1 - \eta)^H$, but gives no positive lower bound on indefinite survival when $\eta > 0$. Finite-horizon persistence and metastability are therefore not exact EGR persistence.

Definition 4.2 (Analytical persistence status). An *exact EGR* satisfies (11). An (H, η) -*quasi-regime* satisfies the stated finite-horizon survival bound but is not called an exact EGR. A *metastable regime extension* must additionally specify a disturbance family and an exit-time scale. These statuses are not interchangeable.

4.3. Descriptors and limiting occupation laws

Let $\mathcal{Y}$ be compact metrizable and let d_{BL} be a bounded-Lipschitz metric that metrises weak convergence on $\Delta(\mathcal{Y})$. Define

$$\widehat{\nu}_T = \frac{1}{T} \sum_{t=0}^{T-1} \delta_{h(X_t)}. \quad (13)$$

A probability law ν is a limiting occupation law under almost-sure weak convergence if

$$d_{\text{BL}}(\widehat{\nu}_T, \nu) \rightarrow 0 \quad \mathbb{P}_{\lambda_0}^{\pi^*}\text{-a.s.} \quad \forall \lambda_0 \in \mathcal{D}(B_0). \quad (14)$$

Analogous definitions apply in probability or in distribution.

The image process $h(X_t)$ need not be Markov. We call h a Markov factor for P^{π^*} when there exists a measurable Markov kernel $P_h^{\pi^*}$ on $\mathcal{Y}$ such that

$$P^{\pi^*}(h^{-1}(C) \mid x) = P_h^{\pi^*}(C \mid h(x)) \quad \forall x \in \mathcal{B}, C \in \mathcal{B}(\mathcal{Y}). \quad (15)$$

The fibre-consistency condition

$$h(x) = h(x') \implies P^{\pi^*}(h^{-1}(C) \mid x) = P^{\pi^*}(h^{-1}(C) \mid x') \quad (16)$$

for every Borel C is necessary and, together with measurable factorisation, sufficient for such a kernel. Only when (15) is available is ν called invariant on the factor space; otherwise it remains a limiting occupation law.

Definition 4.3 (Equilibrium-generated regime). An EGR is a tuple

$$\mathcal{R} = (\pi^*, \mathcal{B}, B_0, h, \nu, \mathfrak{c}) \quad (17)$$

such that: (i) π^* is the selected equilibrium of the complete game; (ii) assumption 4.1 holds; (iii) $\mathcal{B}$ is exactly invariant under P^{π^*} ; and (iv) ν is a limiting occupation law for every initial law in $\mathcal{D}(B_0)$ under the stated convergence mode $\mathfrak{c}$.

Common asymptotic descriptors include fixed points, periodic or recurrent factor dynamics, balanced growth, contraction, and approach to a declared target set. These classes should be defined by explicit topological or probabilistic conditions rather than interpretive adjectives.

4.4. Multi-axis regime profile

An EGR can be described by a multi-axis profile

$$\Xi(\mathcal{R}) = \left(\Xi^A, \Xi^M, \Xi^C, \Xi^R, \Xi^X \right), \quad (18)$$

where the components record asymptotic form, strategic-mode structure, established constitutive relationships, resistance dynamics, and opponent-mediated cross-effects. The axes are analytically distinct but are not assumed to be mathematically independent or mutually exclusive.

Table 2. Illustrative multi-axis regime profile.

Axis	Examples of admissible labels
Asymptotic form	fixed-point, periodic, recurrent, balanced-growth, contracting, target-approaching
Strategic mode	mode-varying, policy-locked
Constitutive findings	policy-constitutive, structurally constitutive, institutionally constitutive, not established
Resistance dynamics	transiently reinforcing, stationary level, undermining, recovery
Opponent-mediated effects	material or incentive; absent, unilateral, or mutual
Optional welfare annotation	improving, neutral, deteriorating, comparative welfare loss

Failure to establish policy constitution does not by itself establish exogeneity. Structural, institutional, and policy constitution can coexist. Welfare is an optional annotation relative to a separate functional and is not an intrinsic component of the EGR definition.

5. Constitutive Regime Formation

5.1. Admissible interventions

The intervention must specify which object changes and which primitives remain fixed.

Definition 5.1 (Policy intervention). A policy intervention $\mathcal{J}_k^\pi$ replaces a predeclared component k of π^* by a feasible measurable rule on a declared set of dates or states while holding (A, u, P) fixed. The intervened policy need not be an equilibrium.

Definition 5.2 (Structural and institutional interventions). A structural intervention $\mathcal{J}^P$ changes a predeclared component of the transition kernel while preserving the required kernel measurability; an institutional intervention $\mathcal{J}^H$ changes a predeclared admission rule, information structure, or feasible-action correspondence. Each protocol must state which remaining primitives are held fixed and must supply an admissible post-intervention behaviour rule whenever the original policy becomes infeasible. These interventions identify constitutive mechanisms, not constitutive equilibrium action.

Let ψ be a predeclared regime-property map on admissible path laws and descriptors, with codomain $(\mathcal{Z}_\psi, d_\psi)$. Examples include exact persistence, growth-rate class, limiting composition, or preserved strategic mode.

Definition 5.3 (Constitutive structural or institutional component). A declared structural or institutional component is constitutive of ψ relative to an admissible intervention $\mathcal{J}$ and a predeclared Borel initial set B_1 containing at least two states with distinct baseline path laws when the baseline path law and the path law under $\mathcal{J}$ differ in ψ for every $x \in B_1$, using the same property map and descriptor on both sides. Uniform constitution requires a positive lower bound on the corresponding d_ψ -distance over B_1 . This classification concerns a mechanism component and does not by itself establish constitution by equilibrium action.

Definition 5.4 (Constitutive policy component). A policy component k is $(\psi, \mathcal{J}_k^\pi, B_1)$ -constitutive if $B_1 \subseteq B_0$ is a predeclared Borel set containing at least two initial states with distinct equilibrium path laws and

$$\psi(\mathbb{P}_x^{\pi^*}) \neq \psi(\mathbb{P}_x^{\mathcal{J}_k^\pi(\pi^*)}) \quad \forall x \in B_1. \quad (19)$$

It is *uniformly constitutive* if there is $\delta_\psi > 0$ such that the corresponding d_ψ -distance is at least δ_ψ for every $x \in B_1$.

Definition 5.5 (Permansson regime). A Permansson regime is an EGR for which at least one predeclared component of the equilibrium policy is constitutive of a defining regime property under an admissible policy intervention.

The terminology is relative to the selected equilibrium, descriptor, property map, and intervention protocol. A process driven by an exogenous trend can be an EGR without being a Permansson regime relative to the growth property.

5.2. Strategic modes and deviation resistance

For each player, let

$$\kappa_i : \text{Gr}(A_i) \rightarrow K_i \quad (20)$$

be a measurable map from feasible state–action pairs to a finite set of strategic modes. An EGR is policy-locked if

$$\kappa_i(x, \pi_i^*(x)) = k_i^* \quad \forall i, x \in \mathcal{B}. \quad (21)$$

Definition 5.6 (Policy-locked Permansson regime). A Permansson regime is policy-locked when its selected equilibrium policy remains within the same predeclared strategic-mode profile throughout the regime region.

Define the set of mode-changing unilateral deviations by

$$\mathfrak{D}_i(x) = \{a_i \in A_i(x) : \kappa_i(x, a_i) \neq \kappa_i(x, \pi_i^*(x))\}. \quad (22)$$

The substantive resistance margin is defined on $\mathcal{X}_i^D = \{x : \mathfrak{D}_i(x) \neq \emptyset\}$:

$$m_i(x) = Q_i^{\pi^*}(x, \pi_i^*(x)) - \max_{a_i \in \mathfrak{D}_i(x)} Q_i^{\pi^*}(x, a_i). \quad (23)$$

Let

$$R_i(x) = \max_{a_i \in A_i(x)} Q_i^{\pi^*}(x, a_i) - \min_{a_i \in A_i(x)} Q_i^{\pi^*}(x, a_i), \quad (24)$$

and, for $x \in \mathcal{X}_i^D$, define

$$\bar{m}_i(x) = \begin{cases} m_i(x)/R_i(x), & R_i(x) > 0, \\ 0, & R_i(x) = 0. \end{cases} \quad (25)$$

For integration and drift bookkeeping, extend $\bar{m}_i$ by zero on $\mathcal{X} \setminus \mathcal{X}_i^D$. Substantive resistance claims are restricted to laws concentrated on $\mathcal{X}_i^D$, so this extension has no behavioural interpretation outside the domain where a mode-changing deviation exists. On $\mathcal{X}_i^D$, the statistic lies in $[0, 1]$ and is invariant to positive-affine transformations of action values.

A horizon- H transient reinforcement claim requires the relevant path law to remain in $\mathcal{X}_i^D$ at dates 0 and H almost surely, and takes the form

$$\mathbb{E}_x^{\pi^*} [\bar{m}_i(X_H) - \bar{m}_i(X_0)] > 0. \quad (26)$$

If the original chain has invariant law μ with $\mu(\mathcal{X}_i^D) = 1$, stationary resistance is the level

$$\mathfrak{R}_i(\mu) = \int \bar{m}_i(x) \mu(dx), \quad (27)$$

not a positive drift. Under an invariant probability law, any integrable diagnostic with non-negative conditional drift almost everywhere must have zero drift almost everywhere. Consequently, perpetual positive reinforcement cannot coexist with stationarity unless positive and negative drift offset across states or the diagnostic is non-integrable.

Comparative-static reinforcement should ordinarily be stated as a finite difference between two fully specified equilibrium selections and invariant laws. A derivative claim additionally requires differentiability of the selected equilibrium, invariant law, mode map, and deviation set.

5.3. Opponent-mediated cross-effects

Let $a_j^0(x) \in A_j(x)$ be a predeclared measurable reference action, and let the one-period intervention replace player j 's date-zero action by $a_j^0(x)$ before returning to π^* . For an integrable diagnostic ϕ_i

and horizon $H \geq 1$, define

$$C_{ij}^{\phi, H}(x) = \mathbb{E}_x^{\pi^*}[\phi_i(X_H)] - \mathbb{E}_x^{j \leftarrow a_j^0}[\phi_i(X_H)]. \quad (28)$$

Player j is strictly cross-generative for player i on B_1 when $C_{ij}^{\phi, H}(x) > 0$ for every $x \in B_1$ and a declared state-support diagnostic ϕ_i . It cross-reinforces player i 's strategic mode when the same condition holds for $\phi_i = \bar{m}_i$ and the margin is defined almost surely under both comparison laws at horizon H . The reference action, horizon, and diagnostic are part of the estimand and should be accompanied by sensitivity analysis over plausible alternatives.

A positive cycle in the cross-effect graph can support feedback, but does not by itself imply persistence, stability, or system-wide escalation.

6. Feedback Architecture and Endogenous Limitation

6.1. Signed feedback graph

For a declared application, let

$$\mathcal{G}_F = (V_F, E_F, s_F, H_F) \quad (29)$$

be a signed, horizon-indexed feedback graph. The nodes in V_F are measurable state diagnostics, the directed edges in E_F are model-relative effects, $s_F : E_F \rightarrow \{-, 0, +\}$ records their sign, and $H_F : E_F \rightarrow \mathbb{N}$ records the evaluation horizon. Typical diagnostics include capability x_i , opponent pressure p_i , internal cohesion c_i , and relative dominance d_i . Every edge must also declare its diagnostic, comparison operation or intervention, baseline law, and evaluation horizon. The graph records a specified comparison structure; it does not by itself establish causal identification.

A positive opponent-mediated edge can represent

$$x_j \longrightarrow p_i \longrightarrow c_i \longrightarrow x_i, \quad (30)$$

while a negative internal edge can represent

$$d_i \longrightarrow c_i. \quad (31)$$

The graph therefore distinguishes direct state cross-generation, incentive reinforcement, counter-mobilisation, and internal limitation rather than assigning all positive feedback to a single reinforcement coefficient.

6.2. Model-relative self-limitation

Let $\mathcal{S}_i^\varepsilon : \mathcal{X} \rightarrow \mathcal{X}$ be a predeclared, measurable state perturbation that increases an own-success or relative-dominance diagnostic for player i by $\varepsilon > 0$ while holding the components specified by the model fixed. For an integrable diagnostic ϕ_i and horizon H , define

$$L_i^{\phi, H}(x; \varepsilon) = \mathbb{E}_{\mathcal{S}_i^\varepsilon x}^{\pi^*}[\phi_i(X_H)] - \mathbb{E}_x^{\pi^*}[\phi_i(X_H)]. \quad (32)$$

The channel is *endogenously self-limiting* relative to $(\phi_i, H, \mathcal{S}_i^\varepsilon)$ on B_1 when $L_i^{\phi, H}(x; \varepsilon) < 0$ for all $x \in B_1$. This is a comparative-dynamics diagnostic, not a policy intervention and not by itself a causal identification theorem. In differentiable special models, (32) may be represented locally by a negative derivative.

The architecture permits asymmetry. One actor may be limited mainly by internal cohesion loss while another is limited mainly by opponent counter-mobilisation, resource congestion, or an institutional response. No symmetry of feedback functions is imposed.

6.3. Co-constitution and co-stabilisation

Definition 6.1 (Antagonistically co-constituted regime). An EGR is antagonistically co-constituted relative to a declared objective-opposition criterion and regime property ψ when: (i) the relevant actors satisfy the opposition criterion; (ii) each actor's equilibrium policy component is constitutive of the same property ψ under its declared admissible policy intervention; and (iii) the signed feedback graph contains a directed interaction cycle connecting their state diagnostics.

Definition 6.2 (Antagonistically co-stabilised regime). An antagonistically co-constituted regime is co-stabilised when it has a declared stability or return property and at least one restorative negative-feedback channel is structurally constitutive of that property under a predeclared structural intervention. Positive cross-effects alone do not establish co-stabilisation.

These definitions separate jointly producing a regime from jointly stabilising it. In the canonical linear model below, each engagement component is constitutive of the same balanced-growth property; the model does not establish objective opposition or restorative co-stabilisation.

7. Scale Dynamics

Let $\varsigma : \mathcal{X} \rightarrow (0, \infty)$ be a declared measurable scale functional satisfying the required log-integrability conditions. For each admissible initial state x , define

$$G_T(\pi^*, x) = \frac{1}{T} \mathbb{E}_x^{\pi^*} \left[\log \frac{\varsigma(X_T)}{\varsigma(x)} \right], \quad (33)$$

with state-indexed lower and upper exponential rates

$$\underline{g}_\varsigma(x) = \liminf_{T \rightarrow \infty} G_T(\pi^*, x), \quad \overline{g}_\varsigma(x) = \limsup_{T \rightarrow \infty} G_T(\pi^*, x). \quad (34)$$

The process is escalatory from x if $\underline{g}_\varsigma(x) > 0$, contracting from x if $\overline{g}_\varsigma(x) < 0$, and exponential-rate neutral from x if both rates are zero. It is *uniformly* escalatory on a declared set B_1 when $\inf_{x \in B_1} \underline{g}_\varsigma(x) > 0$, and uniformly contracting when $\sup_{x \in B_1} \overline{g}_\varsigma(x) < 0$. All remaining cases are rate-indeterminate unless a more specific class is declared. The exact exponent $g_\varsigma(x)$ is used only when the limit exists. Multiplying ς by a positive constant does not change any state-indexed rate.

For $X_t \in \mathbb{R}_+^d$, balanced growth combines $\varsigma(X_t) \rightarrow \infty$ with convergence or occupation convergence of a normalised composition map such as $h(x) = x/\|x\|_1$.

8. Canonical Mutual-Engagement Game

8.1. Primitives

There are two players with capability state $x = (x_1, x_2)^\top \in \mathbb{R}_+^2$, where $\mathbb{R}_+ = [0, \infty)$. Each chooses $a_i \in \{E, W\}$, interpreted as engage or withdraw.

Assumption 8.1 (Canonical parameter class). For $i \neq j$:

(C1) $\ell_i : \mathbb{R}_+ \rightarrow \mathbb{R}$ is bounded, continuous, and non-decreasing.

(C2) $\sigma_i : \mathbb{R}_+ \rightarrow [0, 1]$ is continuous, strictly increasing, satisfies $\sigma_i(0) = 0$, and converges to one.

(C3) $b_i > c_i > 0$.

(C4) $\delta_i \in (0, 1)$ and $\alpha_i > 0$.

(C5) Mutual engagement produces $x' = M_E x$, while every other action profile produces $x' = M_W x$, where

$$M_E = \begin{bmatrix} 1 - \delta_1 & \alpha_1 \\ \alpha_2 & 1 - \delta_2 \end{bmatrix}, \quad M_W = \text{diag}(1 - \delta_1, 1 - \delta_2).$$

(C6) Stage payoff is

$$u_i(x, a) = \ell_i(x_i) + \mathbb{1}\{a_i = E\}(\mathbb{1}\{a_j = E\}b_i\sigma_i(x_j) - c_i).$$

Define thresholds

$$\theta_1 = \sigma_2^{-1}(c_2/b_2), \quad \theta_2 = \sigma_1^{-1}(c_1/b_1), \quad (35)$$

and the engagement cone

$$\mathcal{B} = [\theta_1, \infty) \times [\theta_2, \infty). \quad (36)$$

Consider the threshold policy

$$\pi^*(x) = \begin{cases} (E, E), & x \in \mathcal{B}, \\ (W, W), & x \notin \mathcal{B}. \end{cases} \quad (37)$$

8.2. Invariance and equilibrium

Theorem 8.2 (Invariance and equilibrium in the canonical game). *Under Assumption 8.1:*

(i) $M_E \mathcal{B} \subseteq \mathcal{B}$ if and only if

$$\alpha_1 \theta_2 \geq \delta_1 \theta_1, \quad \alpha_2 \theta_1 \geq \delta_2 \theta_2. \quad (38)$$

(ii) *The threshold policy (37) is a pure stationary MPE of the complete infinite-horizon game for every parameter vector satisfying Assumption 8.1; closure is required for the EGR, not for the equilibrium certificate.*

Proof sketch. The closure conditions are obtained by evaluating the positive linear map at the lower corner (θ_1, θ_2) . Independently of closure, the threshold-policy transition and induced reward are componentwise order-preserving. Inside $\mathcal{B}$, engagement gives a non-negative immediate surplus and a successor state weakly above the withdrawal successor; outside $\mathcal{B}$, unilateral engagement pays the cost without activating mutual-engagement dynamics. The one-shot deviation characterisation therefore establishes MPE throughout the complete game. Full details appear in the technical supplement. $\square$

8.3. Spectral and compositional classification

The Perron eigenvalue is

$$\lambda_+ = 1 - \frac{\delta_1 + \delta_2}{2} + \sqrt{\left(\frac{\delta_1 - \delta_2}{2}\right)^2 + \alpha_1 \alpha_2}. \quad (39)$$

Theorem 8.3 (Canonical spectral classification). *For every admissible parameter vector,*

$$\text{sgn}(\lambda_+ - 1) = \text{sgn}(\alpha_1 \alpha_2 - \delta_1 \delta_2). \quad (40)$$

Under closure, $\lambda_+ \geq 1$. We call closure binding when both inequalities in (38) bind and strict when at least one is strict. Equality $\lambda_+ = 1$ holds exactly under binding closure; strict closure implies $\lambda_+ > 1$.

Theorem 8.4 (Perron compositional convergence). *For every $x_0 \gg 0$,*

$$\frac{M_E^t x_0}{\|M_E^t x_0\|_1} \longrightarrow \frac{v_+}{\|v_+\|_1}, \quad (41)$$

where $v_+ \gg 0$ is the Perron eigenvector. If $\lambda_+ > 1$, $\|M_E^t x_0\|_1 \sim C(x_0) \lambda_+^t$ for a positive constant $C(x_0)$; if $\lambda_+ = 1$, the state converges to a finite point on the Perron ray; if $\lambda_+ < 1$, scale contracts to zero.

For any initial law supported on $\mathcal{B}$, normalised composition converges almost surely to the Perron composition under closure. Together with exact invariance, the predeclared composition descriptor, and the occupation-law theorem in the supplement, strict closure produces a balanced-growth EGR, while binding closure produces a scale-neutral Perron EGR.

8.4. Constitutive engagement and cross-generation

For each i , let $\mathcal{J}_i^W$ force player i to withdraw permanently, holding the payoff and transition primitives fixed. Each is a policy intervention; because every non-mutual-engagement profile activates the same transition rule, the resulting process follows M_W .

Proposition 8.5 (Uniform constitutive dependence of growth). *Under strict closure, equilibrium*

scale has exponent $\log \lambda_+ > 0$, while the permanent-withdrawal intervention has exponent

$$\log \max\{1 - \delta_1, 1 - \delta_2\} < 0. \quad (42)$$

Each player's equilibrium engagement component, and hence mutual engagement, is uniformly constitutive of the growth-sign classification on $\mathcal{B}$ relative to the corresponding permanent-withdrawal intervention.

For $\phi_i(x) = x_i$ and one-period reference withdrawal,

$$C_{12}^{\phi_1,1}(x) = \alpha_1 x_2 > 0, \quad C_{21}^{\phi_2,1}(x) = \alpha_2 x_1 > 0 \quad (43)$$

for every $x \in \mathcal{B}$. The canonical EGR is mutually cross-generative. It is not called cross-incentive reinforcing without an additional proof for the normalised deviation margin.

The canonical model thus supplies a policy-locked, mutually cross-generative, balanced-growth Permansson regime. The mathematical novelty lies in how the benchmark instantiates the regime architecture, not in the positive-matrix results themselves.

Remark 8.6 (Constant-cohesion special case). The linear benchmark deliberately fixes dissipation and suppresses state-dependent cohesion loss, fragmentation, congestion, exhaustion, and other endogenous limits. It isolates reciprocal state generation. The next section embeds it as a constant-cohesion boundary case of a broader non-negative feedback system.

9. Nonlinear Mobilisation and Self-Limitation Extension

Let capability be $x_t = (x_{1,t}, x_{2,t}) \in \mathbb{R}_+^2$ and internal mobilisation or cohesion be $c_t = (c_{1,t}, c_{2,t}) \in [0, 1]^2$. Within the mutual-engagement branch, consider

$$x_{i,t+1} = [1 - \delta_i - \gamma_i(1 - c_{i,t})] x_{i,t} + \alpha_i c_{i,t} x_{j,t}, \quad i \neq j, \quad (44)$$

with

$$c_{i,t+1} = (1 - \rho_i) c_{i,t} + \rho_i F_i(p_i(x_t), d_i(x_t)). \quad (45)$$

Assume $\delta_i \in (0, 1)$, $\gamma_i \geq 0$, $\delta_i + \gamma_i \leq 1$, $\alpha_i > 0$, $\rho_i \in (0, 1]$, the diagnostics $p_i : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ and $d_i : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ are Borel measurable, and $F_i : \mathbb{R}_+ \times \mathbb{R} \rightarrow [0, 1]$ is Borel measurable. In an application, p_i may measure opponent pressure and d_i own relative dominance. When mobilisation and internal limitation are part of the specification, one may additionally impose that F_i is increasing in p_i and decreasing in d_i . These monotonicities are not needed for proposition 9.1 and may differ across actors.

Proposition 9.1 (State-space preservation). *Under the stated restrictions, the update is a Borel measurable self-map of $\mathbb{R}_+^2 \times [0, 1]^2$.*

Proof. Because $\delta_i + \gamma_i \leq 1$ and $c_{i,t} \in [0, 1]$,

$$1 - \delta_i - \gamma_i(1 - c_{i,t}) \geq 1 - \delta_i - \gamma_i \geq 0.$$

Every term in (44) is therefore non-negative. Equation (45) is a convex combination of two numbers in $[0, 1]$, so it also remains in $[0, 1]$. Borel measurability follows from the measurability of p_i , d_i , and F_i and closure of Borel maps under arithmetic composition. $\square$

Proposition 9.2 (Exact nesting of the linear benchmark). *If $c_{i,0} = 1$ and $F_i \equiv 1$ for both players, then $c_{i,t} \equiv 1$ for every t and (44) reduces exactly to $x_{t+1} = M_E x_t$. No restriction beyond the stated bound on γ_i is required because the self-limitation term vanishes at unit cohesion.*

Proof. Substitution into (45) gives $c_{i,t+1} = (1 - \rho_i) + \rho_i = 1$ whenever $c_{i,t} = 1$, so induction gives constant cohesion. With $c_{i,t} = 1$, the term $\gamma_i(1 - c_{i,t})$ vanishes and (44) becomes $(1 - \delta_i)x_{i,t} + \alpha_i x_{j,t}$. $\square$

State-dependent mobilisation and self-limitation allow particular specifications to represent bounded recurrence, endogenous correction, multistability, or basin escape, but none follows from the sign restrictions alone. Existence, bifurcation, stability, equilibrium certification, and constitutive classification for particular nonlinear specifications require separate assumptions and analysis. In particular, Proposition 9.1 is not an MPE theorem and does not establish a nonlinear EGR.

10. Recognition Architecture and Reduced-Form Benchmark

10.1. Belief, latent recognition, public act, and operative classification

Let

$$\mathbf{B}_t \in \mathcal{B}_R \quad \text{belief state,} \quad (46)$$

$$\Lambda_t \in \mathcal{L} \quad \text{latent recognition stance,} \quad (47)$$

$$\mathcal{T}_t \in \mathcal{T}_R \quad \text{submitted text or claim bundle,} \quad (48)$$

$$\mathcal{H}_t \in \mathcal{H}_R \quad \text{operative procedure or admission rule,} \quad (49)$$

$$Y_t \in \mathcal{Y}_R \quad \text{observable vote, declaration, judgment, or classifier output,} \quad (50)$$

$$Z_t \in \mathcal{Z}_R \quad \text{operative institutional classification.} \quad (51)$$

An observation model maps $(\mathbf{B}_t, \Lambda_t, \mathcal{T}_t, \mathcal{H}_t)$ into Y_t , and the admission rule is

$$Z_{t+1} = \text{Adm}(Z_t, Y_t, \mathcal{H}_t). \quad (52)$$

A recogniser may privately accept a proposition, abstain on a bundled public text, and nevertheless preserve an operative sanction or assistance rule. Belief, latent stance, public act, and operative classification therefore require separate measurement and admission models.

For multiple claimants, recognisers, and substantive dimensions, latent recognition can be represented as a field

$$\Lambda_t = (\lambda_{i\ell d,t}), \quad (53)$$

where i indexes claimants or propositions, ℓ recognisers or audiences, and d consequence-relevant dimensions. Observable acts may satisfy

$$Y_{\ell,t} \sim M_{\ell}(\cdot \mid B_t, \Lambda_t, \mathcal{T}_t, \mathcal{H}_t), \quad (54)$$

while aggregation, thresholds, decay, diffusion, agenda formation, issue bundling, and forum selection enter the admission mechanism. Equation (53) is a general architecture, not an estimated latent-variable model or a solved dynamic recognition equilibrium in this paper.

A recognition-mediated transition occurs when $Z_{t+1} \neq Z_t$ and at least one continuation-game primitive or continuation value changes. Persistence can be stated with the finite-horizon or exact-invariance concepts used above. In a richer dynamic model, hysteresis should be defined by evidence-indexed entry and reversal probabilities or thresholds.

Definition 10.1 (Causal-origin capture). Let $\tau_{\text{orig}}^r(\omega)$ be the onset of an event under a predeclared temporal or causal rule r , and let $\widehat{\tau}_t$ be the temporal anchor admitted into the operative institutional state. A candidate causal-origin capture occurs when: (i) a predeclared intervention on strategic claim behaviour changes the distribution of $\widehat{\tau}_t$ or the associated operative classification; and (ii) admission of that anchor changes role assignments, institutional consequences, feasible responses, or continuation-game values. Operative admission can change consequences without changing the underlying event history or establishing the truth of the admitted account.

10.2. Claim-or-wait game

The following game is a reduced-form timing benchmark, not a complete dynamic EGR construction. Two players simultaneously choose claiming C or waiting W . The operative branch is $z \in \{0, 1, 2\}$, with continuation value V_i^z for player i . Claiming costs $K_i > 0$. If only player i claims, it receives its branch with probability $q_i \in [0, 1]$; if both claim, branches 1, 2, 0 occur with probabilities $r_1, r_2, r_0 \geq 0$, where $r_0 + r_1 + r_2 = 1$.

The full payoff matrix is shown in table 3. Define

Table 3. Reduced-form claim-or-wait payoff matrix.

	W_2	C_2
W_1	(V_1^0, V_2^0)	$(q_2 V_1^2 + (1 - q_2) V_1^0, -K_2 + q_2 V_2^2 + (1 - q_2) V_2^0)$
C_1	$(-K_1 + q_1 V_1^1 + (1 - q_1) V_1^0, q_1 V_2^1 + (1 - q_1) V_2^0)$	$(-K_1 + r_1 V_1^1 + r_2 V_1^2 + r_0 V_1^0, -K_2 + r_1 V_2^1 + r_2 V_2^2 + r_0 V_2^0)$

$$g_i = U_i(C, W) - U_i(W, W), \quad (55)$$

$$h_i = U_i(C, C) - U_i(W, C). \quad (56)$$

If the opponent claims with probability p_j , the payoff difference between C and W is

$$D_i(p_j) = (1 - p_j)g_i + p_j h_i. \quad (57)$$

The complete Nash correspondence follows from the sign of $D_i(p_j)$. In particular, (W, W) is an equilibrium iff $g_1, g_2 \leq 0$, and (C, C) is an equilibrium iff $h_1, h_2 \geq 0$. The indifference set for player i is

$$Z_i = \{p_j \in [0, 1] : D_i(p_j) = 0\}. \quad (58)$$

If $g_i h_i < 0$, then Z_i contains the unique interior point

$$p_j^* = \frac{g_i}{g_i - h_i}. \quad (59)$$

If $g_i = h_i = 0$, then $Z_i = [0, 1]$; otherwise Z_i has no interior point. Hence a fully interior equilibrium exists exactly when, for each player, either $g_i h_i < 0$ or $g_i = h_i = 0$. It is unique exactly when both products are negative. Boundary and degenerate equilibrium sets are covered by the same best-response condition and are stated completely in the technical supplement.

When $g_i < 0 < h_i$ for both players, both (W, W) and (C, C) are strict equilibria. If both players prefer (W, W) , the game exhibits a defensive-claim coordination problem; a trap interpretation requires an equilibrium-selection or dynamic-attraction mechanism not supplied by the static benchmark.

10.3. Order premium

Let $V_i^{i,F}$ and $V_i^{i,L}$ denote player i 's continuation values when its branch is recognised first and later, holding truth state, evidence, claimant type, and institutional composition fixed. Define

$$\Delta_i^F = V_i^{i,F} - V_i^{i,L}. \quad (60)$$

Holding the remaining continuation values and admission probabilities fixed is a ceteris-paribus comparative static. Under this decomposition,

$$g_i = g_i^L + q_i \Delta_i^F, \quad h_i = h_i^L + r_i \Delta_i^F. \quad (61)$$

Thus the order premium moves claim incentives linearly and is causally decisive for a best response only when it crosses a sign boundary.

Corollary 10.2 (Conditional order-induced branch pre-emption). *Suppose operative classification changes a continuation-game primitive or value, the recognised branch is persistent over the relevant horizon, and $\Delta_i^F > 0$ after the stated controls. If $g_i^L < 0 \leq g_i$, claiming enters player i 's best-response correspondence against a waiting opponent; if $h_i^L < 0 \leq h_i$, it enters against a claiming opponent. Strict positivity makes claiming the unique best response in the corresponding comparison. The resulting Nash equilibria are then determined by the complete sign characterisation of the claim game.*

The first player whose claim is admitted into the operative state may therefore obtain an advantage in determining which continuation-game branch governs subsequent play. The result is conditional. It establishes neither the truth of the first claim nor equal moral responsibility among actors.

Remark 10.3 (First-state-transition advantage). The mechanism is not merely an advantage from acting first within a fixed game. When admission changes Z_{t+1} , the first effective claim can influence which continuation-game branch, payoff structure, or institutional rule governs later play:

$$\text{first move within } \Gamma \neq \text{first effective move in selecting } \Gamma(Z_{t+1}).$$

The remark applies only under the hypotheses of corollary 10.2; order has no strategic effect unless the premium changes a best-response comparison.

11. Welfare, Identification, and Falsification

11.1. Decision loci and stakeholder welfare

The modelled player must be the actual decision locus. Let N index decision-making actors and let S be a separate stakeholder set. A measurable one-period stakeholder-welfare function

$$w : \text{Gr}(A) \rightarrow \mathbb{R} \tag{62}$$

need not coincide with any player's utility. Longer-run evaluation requires a separately declared aggregation functional $\mathfrak{W}$ on admissible path laws. Finite-horizon deterioration, asymptotic welfare loss within one regime, and comparative welfare across regimes are therefore distinct claims. Equilibrium rationality for leadership, firms, agencies, or factions does not imply stakeholder welfare.

Reciprocal strategic effects also do not imply symmetric causation, factual support, freedom of action, or moral responsibility. PRT classifies strategic and causal structure; normative conclusions require separate premises.

11.2. Empirical identification

A credible application must:

1. identify the actual decision-makers and payoff-relevant state;
2. establish or estimate the relevant equilibrium policy and state transition law;
3. state the equilibrium-selection rule when multiplicity is possible;
4. predeclare the regime region, basin, descriptor, and convergence criterion;
5. test exact invariance or explicitly use a finite-horizon quasi-regime;
6. specify an admissible policy intervention and the primitives held fixed;
7. estimate the regime property under equilibrium and intervention laws;
8. define strategic modes, deviation margins, reference actions, and cross-effect sensitivity;

9. declare the signed feedback graph, state-perturbation semantics, and asymmetry of limiting mechanisms when those claims are made;
10. separate player utility from stakeholder welfare.

Recognition applications additionally require separate measurement models for beliefs, latent stance, public acts, and operative classification; explicit treatment of proposal text, abstention, bundling, and procedure; estimates of branch persistence; and a design identifying order independently of evidence and claimant quality. A causal-origin application must declare the onset rule r separately from the operative anchor admitted by the institution and identify the effect of claim behaviour on admission rather than infer capture from temporal disagreement alone.

Structural estimation of dynamic games is demanding, and identified counterfactuals can depend on information restrictions, equilibrium beliefs, and selection assumptions (Bajari et al. 2007; Pesendorfer and Schmidt-Dengler 2008; Aguirregabiria and Magesan 2020; Magnolfi and Roncoroni 2023; Berry and Compiani 2023). A PRT application should therefore report partial identification or sensitivity sets when point identification is not defensible.

11.3. Diagnostic exclusion criteria

A proposed core application fails if:

1. the policy lacks the stated equilibrium certificate;
2. the region or descriptor is selected from the same realised trajectory used for confirmation;
3. the alleged exact region can be exited with positive probability;
4. the limiting law lacks a stated initial class or convergence mode;
5. the policy intervention is not admissible or leaves the claimed property unchanged;
6. cross-generation vanishes under plausible reference actions;
7. an incentive-reinforcement claim disappears under the specified normalisation;
8. recognition does not alter an operative state or continuation value;
9. an order effect disappears after the required controls;
10. a purported self-limitation effect depends on an undeclared or infeasible state perturbation;
11. co-stabilisation is inferred from positive cross-effects without a restorative channel and stability test;
12. a welfare or moral conclusion is inferred directly from equilibrium structure.

Approximate equilibrium, finite-horizon persistence, and metastability are valid extension objects, but should not be inserted into the exact core without restating the solution and persistence concepts.

12. Generalisation Map and Limitations

Table 4. Generalisation map and additional consistency requirements.

Core assumption	Generalisation	Additional consistency requirement
Full observation	Belief-mediated regime	Sequentially consistent equilibrium and belief update
Human or passive recogniser	Algorithmic adjudicator	Latency, adversarial input, chattering, and audit model
Fixed policy	Learning or update rule	Adaptive-state definition and convergence or recurrence concept
Known physical kernel	Robust or ambiguous transition	Separation of physical law from subjective worst-case kernel
Exact persistence	Metastability	Disturbance family, exit-time law, and scale separation
Fixed interaction graph	Endogenous topology	Network-formation incentives and transition consistency
Costless execution	Implementation burden	Feasibility, maintenance cost, and operational failure state
Passive data	Performative recognition	Endogenous data distribution and recogniser feedback

The present paper does not prove existence or uniqueness of the selected equilibrium in the general environment. It does not provide a general statistical identification theorem for the analyst-selected regime specification. It does not show that the canonical model is cross-incentive reinforcing. The nonlinear mobilisation model has only a state-space preservation result; it is not supplied with a general equilibrium, attractor, or bifurcation theorem. The distributed-recognition representation is architectural, and the claim-or-wait benchmark does not establish a dynamic recognition EGR. Constitutive dependence remains relative to a declared intervention family, and self-limitation remains relative to a declared state perturbation and diagnostic.

Approximate equilibrium, finite-horizon persistence, metastability, robust control, performative recognition, endogenous topology, and co-adaptive learning are valid extension objects only after their solution, state, and persistence concepts are restated. They should not be inserted into the exact EGR core by analogy alone.

13. Conclusion

A dynamic equilibrium certifies a policy and induces a state process. This paper formalises a post-equilibrium object for classifying that process. An EGR combines a selected equilibrium with a predeclared region, initial basin, exact persistence, descriptor, limiting occupation law, and convergence mode. A Hermansson regime adds a fixed policy intervention that changes a defining regime property.

The multi-axis profile prevents asymptotic form, policy locking, constitutive dependence, resistance, cross-effects, and welfare from being treated as synonyms. The signed feedback

architecture further distinguishes opponent-mediated state generation from incentive reinforcement and from internal limitation. The canonical engagement game proves the minimal linear case: a policy-locked, mutually cross-generative, balanced-growth Hermansson regime. The nonlinear extension shows how mobilisation and cohesion can enrich the state while preserving non-negativity and nesting the linear benchmark, but leaves attractor and equilibrium results open.

The recognition architecture separates belief, latent stance, observed action, and operative classification. Its reduced-form benchmark shows how an order premium can alter best responses and thereby create a first-state-transition advantage over the continuation branch. Distributed recognition and causal-origin capture are defined as extensions of the same operative-state architecture rather than treated as already solved dynamic games.

The central contribution is therefore not the proposition that actions change states. It is a disciplined method for asking which regime a certified equilibrium generates, whether equilibrium action is constitutive of that regime, how opponents generate material or incentive support, and which positive and negative feedback channels sustain, limit, redirect, or dissolve the interaction.

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A. Notation Summary

Symbol	Meaning
Γ	Discounted stochastic game.
$\mathcal{X}$	Complete payoff-relevant Polish state space.
$A_i(x)$	Player i 's feasible actions.
π^*	Selected equilibrium policy.
P^{π^*}	Equilibrium-induced transition kernel.
$\mathcal{B}$	Predeclared regime region.
B_0	Initial-state basin.
$\mathcal{D}(B_0)$	Initial laws supported on B_0 .
h	Regime descriptor or normalisation map.
ν	Limiting occupation law.
$\mathcal{R}$	Equilibrium-generated regime.
$\mathcal{J}_k^\pi$	Policy intervention on component k .
ψ	Predeclared regime-property map.
κ_i	Strategic-mode map on the feasible state–action graph.

Symbol	Meaning
$\mathfrak{D}_i(x)$	Mode-changing unilateral deviations.
$\bar{m}_i(x)$	Normalised mode-resistance margin.
$C_{ij}^{\phi, H}$	Horizon- H cross-effect of player j on diagnostic ϕ_i .
$\underline{g}_\varsigma, \bar{g}_\varsigma$	Lower and upper exponential scale-growth rates.
$\Xi(\mathcal{R})$	Multi-axis regime profile.
$\mathcal{G}_F$	Signed, horizon-indexed feedback graph.
$L_i^{\phi, H}$	Model-relative self-limitation diagnostic.
(x_t, c_t)	Capability and cohesion state in the nonlinear extension.
$\mathbf{B}_t, \Lambda_t, \mathcal{T}_t, \mathcal{H}_t, Y_t, Z_t$	Belief, latent stance, submitted claim bundle, operative procedure, observed act, and operative classification.
$\tau_{\text{orig}}^r, \widehat{\tau}_t$	Rule-indexed event onset and operative temporal anchor.
$w, \mathfrak{W}$	One-period stakeholder welfare and path-law aggregation functional.
g_i, h_i	Claim gains against a waiting and claiming opponent.

B. Formal Claim Boundary

The paper proves the general one-shot deviation characterisation, exact-invariance equivalence and finite-horizon bound, point-convergence implication for occupation measures, positive-affine invariance of the normalised resistance margin, stationary no-drift result, and multiplicative-unit invariance of the scale rates. For the canonical parameter class it proves MPE of the threshold policy without a closure assumption, exact cone closure, spectral classification, Perron compositional convergence, constitutive growth dependence, and mutual material cross-generation. For the nonlinear mobilisation extension it proves only preservation of $\mathbb{R}_+^2 \times [0, 1]^2$ and exact nesting of the linear benchmark under the stated restriction. For the reduced-form claim game it proves the complete Nash correspondence including pure, boundary, interior, and degenerate cases, defensive-claim coordination result, and the ceteris-paribus effect of the order premium on claim incentives.

The paper does not prove general pure-MPE existence, empirical identification, canonical cross-incentive reinforcement, nonlinear equilibrium or attractor existence, restorative co-stabilisation, a dynamic recognition EGR, or the deferred incomplete-information, robust, performative, spatial, metastable, and learning extensions. The distributed-recognition field, causal-origin capture, signed feedback graph, and multi-axis profile are formal architectures or diagnostics whose empirical and dynamic content must be established in each application.

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